



Cockpit Platform Development

Company Name

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Course: M.Sc Software Systems



Need for the system:

In complex software systems, diagnosing runtime failures requires manually correlating unstructured error logs with source code across potentially thousands of files and functions. This process is time-consuming, error-prone, and demands deep codebase familiarity. There is a critical need for an automated system that can transform raw runtime logs into precise, code-grounded root-cause analysis with actionable fix recommendations — bridging the gap between runtime observations and static code understanding.



Problem Objective

1. **Automated Log-to-Code Mapping** - Transform raw runtime error logs into precise source code locations (file, function, line range) without manual intervention.
2. **Semantic Code Analysis** - Leverage Code Property Graphs (CPG) to capture code semantics — control flow, data flow, and call relationships — enabling queries that pure text search cannot achieve.
3. **AI-Enhanced Root-Cause Reasoning** - Integrate an LLM-based agent with on-demand tool calling to perform deep, iterative code analysis and produce structured root-cause findings with fix recommendations.
4. **Multi-Component Support** - Analyze logs referencing multiple repositories/components in a single run, loading each CPG once and reusing across all log entries.
5. **Structured, Actionable Output** - Generate machine-readable JSON output with root-cause categories, failure mechanisms, and ranked fix options — ready for downstream automation.



- **Static Code Analysis** — Parse C/C++ source code into Code Property Graphs (CPG) for semantic understanding of control flow, data flow, and call relationships.
- **Runtime Log Diagnostics** — Map runtime error logs (raw text or structured JSON) to precise source code locations using CPGQL graph queries against a local Joern server.
- **AI-Powered Root-Cause Analysis** — LLM agent with 8 built-in tools and iterative tool-calling loop for on-demand context retrieval, producing structured root-cause findings with ranked fix recommendations.
- **Multi-Component Batch Processing**— Analyze logs from multiple repositories in a single run — load each CPG once and reuse across all log entries without server restart.
- **Structured Diagnostic Output** — Generate machine-readable JSON reports (source locations, root-cause categories, fix options, metrics) and optional human-readable Markdown summaries.

Component	Technology
Static Analysis	Joern / Code Property Graph
Graph Storage	OverflowDB (binary)
Query Language	CPGQL
LLM	OpenAI-compatible API
Language	Python 3.8+



Requirement Analysis

- **Automated Log Analysis** — System that automatically processes raw error logs and identifies the exact source code location responsible for each error, eliminating manual log-to-code correlation
- **Intelligent Root-Cause Reasoning** — AI-assisted analysis that categorizes the defect type and explains the failure mechanism step-by-step
- **Actionable Fix Recommendations** — Ranked fix options with code snippets that conform to the existing codebase, not generic suggestions
- **Multi-Repository Support** — Analyze logs from multiple components/repositories in a single run without restarting the analysis server
- **Real-Time Feedback** — Streaming output during long-running LLM analyses to monitor progress and detect issues early

Conclusion



- The system automates the identification of root causes from runtime error logs.
- It combines Code Property Graphs with LLM-based reasoning to perform deeper code analysis.
- It provides actionable fix recommendations along with structured analysis results.
- It makes the debugging process faster, systematic, and less dependent on manual investigation.